

Timing of Resistance Training During Ramadan Fasting and Its Effects on Muscle Strength and Hypertrophy

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Purpose: During Ramadan month, Muslims often continue training to maintain their muscle performance. However, trainers should stress the importance of the time of day chosen for practicing resistance training (RT) during Ramadan fasting to avoid health and performance alteration. Thus, this study aimed to investigate the effects of the timing of RT during Ramadan intermittent fasting (RIF) on muscle strength and hypertrophy in healthy male adults. **Methods:** Forty men were randomly allocated to 2 matched groups: 20 practicing whole-body RT in the late afternoon in a fasted state and 20 training in the late evening in a fed state (FED). Both groups performed 4 days per week of whole-body RT (~75%–85% 1-repetition maximum, 12 repetitions/3–4 sets) during RIF. Anthropometric measurements and 1-repetition-maximum test for the squats, deadlift, and bench press were measured. A cross-sectional area of the quadriceps and biceps brachii was examined using ultrasound scans. All measurements were taken at 4 time points: 24 hours before the start of Ramadan, on the 15th day of Ramadan, on the 29th day of Ramadan, and 21 days after Ramadan. **Results:** Post hoc tests indicated significant preimprovement to postimprovement in FED at the 29th day of Ramadan for the 1-repetition-maximum test for the squats ($P = .02$; effect size = 0.21) and deadlift ($P = .03$; effect size = 0.24) when compared to 24 hours before the start of Ramadan. No significant changes were observed in the fasted-state group or for cross-sectional area for both groups. **Conclusions:** Practicing RT during RIF appears not to have adverse effects on muscle hypertrophy and strength no matter whether it is practiced in FED or fasted state. However, RT had greater effects on muscle strength when applied in FED. Our findings suggest that training sessions should be scheduled after breaking the fast during Ramadan.

Keywords: males, exercise training, adaptation, energy-intake timing

Ramadan fasting is an Islamic religious obligation requiring healthy pubertal Muslim adolescents and adults to fast from dawn to sunset for approximately 28 to 30 consecutive days.¹ This fasting involves abstaining from food, fluid intake, engaging in sexual activities, and smoking for ~12 to 20 hours per day, depending on the geographic location and season of the year.²

Changes in mealtimes and the consumption of food and drink are unrestricted only during the hours of darkness.³ Caloric restriction during the Ramadan intermittent fasting (RIF) decreases total cholesterol, low-density lipoprotein, fasting blood glucose levels,⁴ and also improves body composition (eg, lower body fat and higher lean body mass),⁵ possibly by mobilizing saturated fatty acids for metabolic processes.⁶ However, reductions in total energy and water intake, less

daily meals (from 3–4 to 2 meals—a late evening meal and/or an early breakfast), and changes in diet composition can induce acute diurnal dehydration,⁷ impair sleep quality,^{8,9} reduce total protein synthesis,¹⁰ and create low energy availability.¹¹ These changes in terms of food intake and sleep can impact on physiological and psychological functions and may specifically impose major challenges for Muslim athletes who continue training and competitions while fasting.³

Abstaining from eating and drinking for many hours before and during exercise can decrease the availability and utilization of energetic substrates and also cause changes in hormonal and metabolic responses in parallel with dehydration.^{12,13} In addition, sleep deprivation and feelings of malaise increase in fatigue and perceived efforts in response to the same training volume compared to the non-Ramadan period.^{14,15} Most athletes experience RIF with a limited loss of physical performance during aerobic^{16,17} and anaerobic exercises.^{18,19}

Resistance training (RT) is an exercise modality commonly used to improve muscular fitness (ie, muscle strength, power, and local muscle endurance) as well as motor skill development in various types of sports including soccer²⁰ and martial arts.²¹ It has further become an increasingly popular mode of physical exercise designed to improve body image and self-rated physical attractiveness by stimulating muscular fitness and muscle cross-sectional area (CSA).^{22,23} Studies of the combined effects of RIF and RT on body composition, muscle hypertrophy, and physiological

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adaptation have produced contradictory findings.^{24,25} For example, while muscle strength and power increased significantly in soccer players after RT (8 repetitions $\times$ 4 sets with 4-min recovery in between) during Ramadan,²⁶ an absence of changes in body mass and body composition have been reported for bodybuilders.²⁷ Moreover, body mass and physical performance declined in male Senegalese runners and decreased after RT due to hypoglycemia associated with RIF.²⁸

These findings suggest that strength and conditioning specialists should choose an appropriate training period to minimize performance degradation and improve/maintain physical outcomes associated with RIF. Some investigators suggest training would be more beneficial after breaking fast (usually 2–3 h after sunset) to prevent dehydration, hypoglycemia, and possible decreases in performance,^{29,30} while other studies suggest that training should take place while fasting to improve sleep quality and prevent hormonal disturbances that can alter performance and muscle growth.³¹

Due to contrasting opinions on the timing of RT during RIF, we investigated the effects of RT on strength and muscle hypertrophy in both fasted (FAST; training before breaking fast) and fed states (FED; training after breaking fast). We hypothesized that (1) RT can be safely practiced during Ramadan in both FED and FAST while maintaining muscle strength and hypertrophy, and (2) there are larger increases in muscle strength and hypertrophy when training in the FED compared with the FAST.

Materials and Methods

Participants

Forty healthy Muslim recreational male weight lifters volunteered to participate in this study (age = 25.7 [5.6] y, mass = 85.1 [17.5] kg, height = 175.3 [8.9] cm, body mass index = 28.3 [5.7] kg/m²) and were randomly allocated to 2 groups: (1) FAST group (n = 20) practicing RT in the late afternoon 1 hour before breaking fast (between 16 and 18 h), and (2) a FED group (n = 20) practicing RT in the late evening after breaking the fast (between 20 and 22 h). The study inclusion criteria were healthy males aged 18–35 years, 2 years of RT experience, free from any injuries that could impede or hinder the movements performed, no current or history of cardiometabolic disease, no consumption of supplements other than those provided during the study (see below), and self-reported as not having previously used anabolic steroids or related ergogenic aids. The participants were required to refrain from other nonexperimental RT sessions during the study. The participant who did not meet the inclusion criteria was excluded.

All participants were informed of the purpose of the study, the experimental procedures, possible discomforts, risks, and benefits of the participation and then signed a written informed consent to participate in the study. The investigation was performed according to the principles outlined in the Declaration of Helsinki and was approved by the local University Ethics Committee on Human Research of the University of la Manouba, Tunisia (Ethics Code: Tn, UM2019-73).

Design

Ramadan began on March 3 and ended on April 3, 2021, with an average duration of the fast being ~14 hours per day. The study was conducted in Tunisia, where daytime temperatures were 23 (4) °C

and relative humidity was 65% (5%). Two weeks before the start of Ramadan fasting, volunteers visited the laboratory on 3 days separated by intervals of 48 to 72 hours between visits.

Visit 1: Participants signed an informed consent form, answered the physical activity readiness questionnaire and the Pittsburgh Sleep Quality Index (PSQI), and underwent anthropometric measurements and 1-repetition-maximum (1RM) tests.

Visit 2: Participants repeated the 1RM tests to determine the loads to be used for the experimental conditions and to assess the reproducibility of the tests.

Visit 3: Ultrasound images were recorded to determine the CSAs of the quadriceps muscle and biceps brachii muscle.

After the baseline assessment (T0), all tests (anthropometrics, 1RM, CSAs, and PSQI) were conducted on 3 separate test occasions: during the second week of Ramadan (T1), during the fourth week of Ramadan (T2), and 21 days after the end of Ramadan (T3), with the same order and the same separated days as the previous pretest procedures (Figure 1).

Training Program

All participants trained 4 days per week (Monday, Tuesday, Thursday, and Friday) for 8 weeks (1 wk before Ramadan, 4 wk during Ramadan, and 3 wk after the end of Ramadan). Training involved full-body RT exercises of 45 to 60 minutes per session. The exercise sessions of FAST occurred in the late afternoon during Ramadan (between 16 and 18 h), and those for the FED occurred in the evening (between 20 and 22 h) at 1 to 2 hours after breaking the fast. All sessions started with a general warm-up and cooldown periods of 5 to 10 minutes of low-intensity aerobic and dynamic stretching exercises.

Five RT exercises were performed in each session, each consisting of 4 sets. Time under tension was 2 seconds in concentric mode and 2 seconds in eccentric mode at an intensity of 75% to 85% of the 1RM. Two minutes of rest was granted between sets and exercises. Each protocol was repeated twice weekly (Mondays/Thursdays: inclined leg press, parallel squat, deadlifts, calf raise, and cable crunches; and Tuesdays/Fridays: bench press, shoulder press, lateral pulldown, barbell curl, and obliques raises).

During the first session of each intervention week, 1RM tests were performed for all exercises to ensure a relative exercise intensity of 75% to 85% 1RM throughout the 8 weeks of training. The 1RM test and the start of the training session were separated by a 10-minute rest interval.

Training protocols were started by performing 3 sets of 12 repetitions at 75% of the 1RM during the first week and progressed to 85% of the 1RM during week 6. This progressive overload was monitored by a trained researcher who supervised each workout session to ensure exercises were performed with the correct technique during the concentric and eccentric phases of the movement. The trained researcher also recorded the compliance and individual workout data during each training session so that exercise prescriptions were properly administered (eg, number of repetitions and sets, rest, velocity of movement, intensity, and respiration).

Anthropometric Measurements

Body composition was assessed following an overnight fast (12 h) and >72 h following their last exercise bouts at T0, T1, T2, and T3. Body mass was measured to the nearest 10 g using a calibrated electronic scale (MEDISANA), and body height was measured to

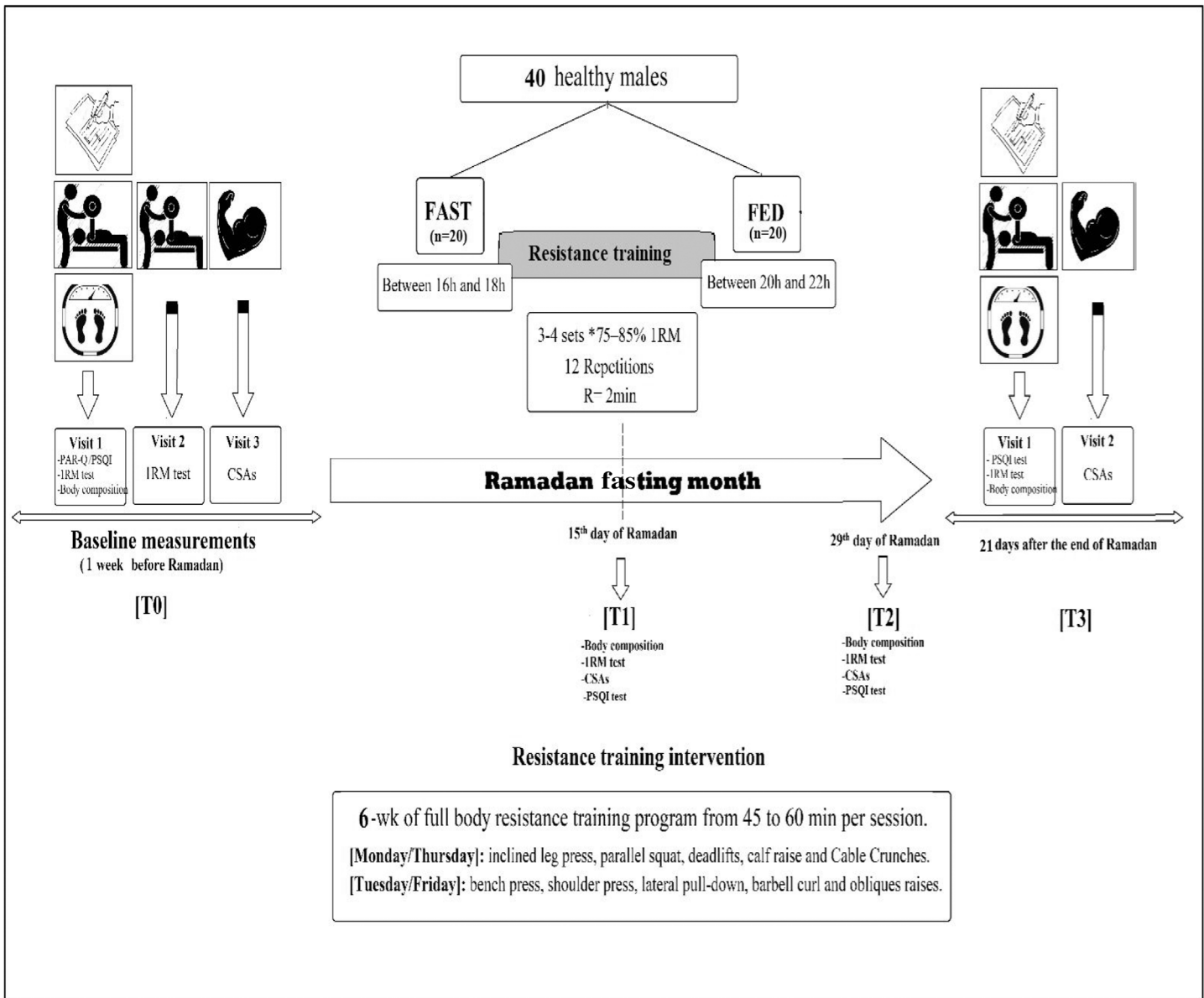


Figure 1 — Study design. 1RM indicates 1-repetition maximum; CSA, cross-sectional area; FAST, fasted state; FED, fed state; PAR-Q, Physical Activity Readiness Questionnaire; PSQI, Pittsburgh Sleep Quality Index; T0, 1 week before the start of Ramadan; T1, the 15th day of Ramadan; T2, the day after the end of Ramadan; T3, 21 days after the end of Ramadan.

the nearest 5 mm using a stadiometer. Body mass index was then calculated. In addition, 4 skinfolds (biceps, triceps, subscapular, and suprailium) were measured with a calibrated Harpenden skinfold (Baty International) for the evaluation of body fat percentage using a previously published algorithm.³² Lean body mass was calculated as body mass minus body fat mass. All measurements were performed by a single evaluator who was experienced in each of the procedures.

Maximum Strength Tests

Maximum strength (1RM tests for parallel squat, bench press, and deadlift) was performed at T0, T1, T2, and T3 and before the first weekly training session to maintain the relative intensity of each exercise. Participants completed 1RM testing using inclined leg press (Hammer Strength), Smith machine squat (Hammer Strength),

standing calf raise machine (Hammer Strength), cable crunches (Hammer Strength), machine-guided shoulder press (Hammer Strength), pulldown machine (Hammer Strength), deadlifts, barbell bench press, and barbell curl using a 20-kg Olympic barbell with free weights.

Participants performed 10 repetitions without any extra load on the equipment as described by others.^{33,34} Values for 1RM were determined in a concentric mode with a maximum of 6 attempts, with 5-minute rest periods between each attempt.³⁴ In addition, a 3-minute rest period was allowed between tests of each exercise.

Pilot data were obtained on 2 test days from 40 participants to determine the reproducibility of the test. Our reliability data revealed the following intraclass correlation coefficients at 95%: 1RM parallel squat (.929; lower .929–upper .996), 1RM bench press (.779; lower .581–upper .883), 1RM deadlift (.886; lower .783–upper .940), 1RM CSA quadriceps (.967; lower .938–upper .983), and 1RM CSA biceps (.782; lower .587–upper .885).

Muscle CSAs

Ultrasound images were recorded on 4 different occasions (before Ramadan [T0], on the 15th day of Ramadan [T1], on the 29th day of Ramadan [T2], and T3) to analyze the CSA of the quadriceps and biceps brachii muscle. Ultrasound scans were performed using a brightness mode imaging device (MyLab X8, Esaote) with a multifrequency linear array probe (4–15 MHz, 60-mm width) using extended field-of-view mode (EFOV).

CSA images of the quadriceps were taken using the EFOV mode.³⁵ Volunteers remained lying in dorsal decubitus position on a stretcher for 15 minutes to allow fluid shifts to stabilize before images were obtained.³⁶ A line from the greater trochanter to the outer femoral condyle was marked to measure the length of the femur. Four axial perpendicular lines were then marked, starting from the proximal extremity at intervals of 20% (proximal), 40% (proximal middle), 60% (distal middle), and 80% (distal) of the femur length using a tape measure and a pachymeter positioned parallel to the thigh (Figure 2A). Lines created with a microporous adhesive tape were then positioned 2 cm from each of the 4 axial perpendicular points to delineate the location for positioning of the ultrasound probe during image acquisition (Figure 2B). It is important to note that CSA measured at a single point on a muscle's length may not adequately represent the entire extent of muscle hypertrophy; thus, CSA analysis using several points provides a more accurate depiction of muscle hypertrophy.³⁷

The probe was placed transversely on the marked lines across the thigh to allow for a continuous single view, while the mediolateral angle of the probe was changed throughout the experiment so that the probe remained perpendicular to the skin (Figure 2C).³⁸ This procedure was performed at constant speed and lasted approximately 4 seconds in distal regions to approximately 5 seconds in proximal regions according to the thigh circumference of the participants.

A water-based gel was applied between the skin and the transducer to achieve acoustic coupling and avoid any misinterpretation of the images created by the increased pressure when applying the transducer. Participants were asked to remain relaxed during the ultrasound procedures. The panoramic images of the 4 sites that were created by the moving transducer were digitally stored (Figure 3). Images were analyzed for muscle thickness, fascicle angle, and fascicle length by using specific software (MyLabDesk, Esaote).

The sum of 4 images of the CSAs was taken at 20%, 40%, 60%, and 80% of femur length on the test day and used to generate

a single CSA value per muscle, which was then used for determining the thickness of the rectus femoris and vastus lateralis muscles.

The CSA of the biceps brachii was also measured by the EFOV technique with the same device and on the same testing day when images of the quadriceps (T0, T1, T2, and T3) were obtained.³⁹ Each individual was seated in a comfortable position with arms resting at 90° shoulder flexion angle and 0° elbow flexion (ie, straight arm). A line from the acromion process of the clavicle to the lateral epicondyle of the humerus was marked to measure the length of the upper arm. Two axial perpendicular lines were marked (midpoint of the upper arm: 0 and 3 cm and below the midpoint: 3 cm) as these 2 regions provide the largest biceps brachii CSA that generally lies between the 2 perpendicular markings.

EFOV scans were obtained by moving the probe along the 2 marked lines axially from the medial aspect to the lateral aspect of the upper arm in a continuous single view. Pressure was applied minimally but consistently, avoiding compression of the muscle, with transmission gel applied between the skin and the transducer to aid in acoustic coupling. Images of the biceps brachii taken from each region were traced to calculate CSA using a computer software program (MyLabDesk, Esaote).

The sum of the 2 CSAs taken at the midpoint of the upper arm was calculated on each testing day: 0 and 3 cm and below the midpoint: 3 cm to generate a single CSA value per muscle, which was then used for data analysis of the biceps brachii muscle thickness.

Dietary Habits

Participants were advised to maintain instructed nutritional regimen by a specialist and to avoid taking any supplements other than those provided during the study (see below). Dietary adherence was assessed by self-reported food records and analyzed using a nutritional tracking application (<http://www.myfitnesspal.com>) and the food composition tables of the National Institute of Statistics of Tunis (1978). All food and beverages consumed during the experiment were collected for at least 4 days per week to offer a reliable estimate of food intake for the entire week. Total water intake was defined as the fluid volume of consumed beverages plus the water content of consumed foods.

Individuals were instructed on the correct recording procedures of all food items, including portion sizes. Each item of food was individually entered into the software program to determine total energy consumption, and amount of energy derived from

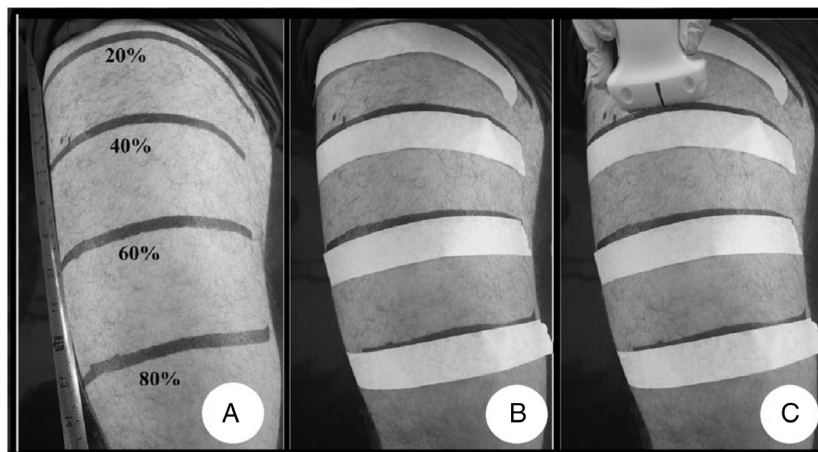


Figure 2 — Thigh-marking procedures (A and B) and ultrasound image acquisition (C).

proteins, fats, and carbohydrates for each period analyzed. The mean of nutrient consumed during the intervention was calculated.⁴⁰ Participants consumed a supplement containing 24-g protein and 1-g carbohydrate before sleeping (Iso100 Hydrolyzed Whey Protein Isolate, Dymatize Nutrition) on training days under the supervision of the research staff. The nutrients consumed by participants during the study are shown in Table 1, and no significant differences between groups and during the experiment were found.

Rating of Perceived Exertion

The Borg scale was used to evaluate the rating of perceived exertion (RPE) during the exercise sessions at 5 minutes after the end of the training session at T0, T1, T2, and T3.⁴¹ Volunteers

were advised to look at the scale and respond immediately after the end of each session to the prompt, “Think about your feeling of body effort, how did you feel?” and then enter numbers on a sliding scale where perceived efforts were rated between 0 for “extremely easy” and 10 for “extremely difficult.”

Sleep Quality

Sleep quality during and after 3 weeks of Ramadan was assessed using the full PSQI⁴² before Ramadan (T0), on the 15th day of Ramadan (T1), on the 29th day of Ramadan (T2), and T3. We used an Arabic validated version of the PSQI.⁴³ The global PSQI score is composed of 7 components: subjective sleep quality (sleep quality perception), sleep latency (time between lying on bed and start

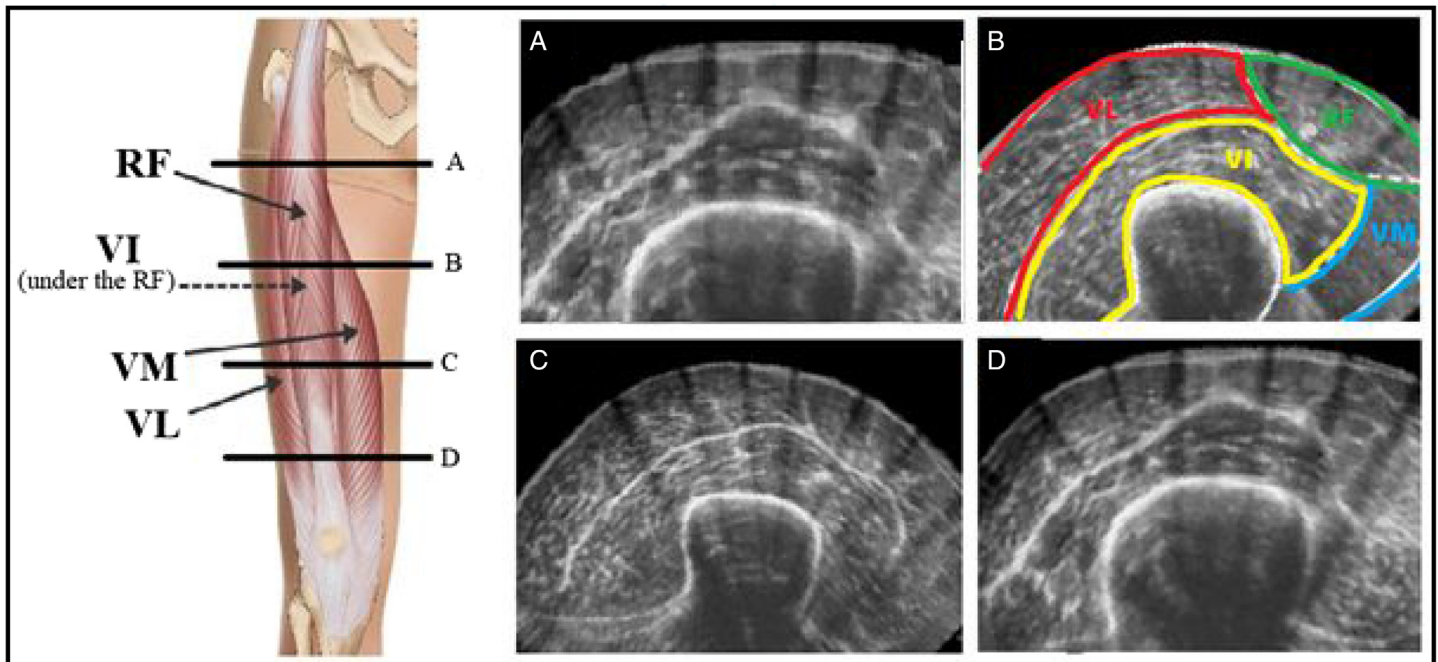


Figure 3 — Ultrasound images and cross-sectional areas of quadriceps at 20% proximal (A), 40% (B), 60% (C), and 80% (D) of femur length. RF indicates rectus femoris; VI, vastus intermedius; VL, vastus lateralis; VM, vastus medialis.

Table 1 Estimated Daily Dietary Intake Recorded of the FAST and FED Groups Measured During T0, T1, T2, and T3 (n = 20 per Group)

Variable	Group	Phase, mean (SD)				P (ES)		
		T0	T1	T2	T3	Time	Group	Group × time
Protein, %kcal	FAST	18 (8)	19 (1)	18 (7)	18 (7)	.08 (0.37)	.06 (0.45)	.10 (0.57)
	FED	18 (6)	18 (2)	18 (9)	18 (6)			
Carbohydrate, %kcal	FAST	47 (5)	45 (8)	46 (3)	47 (1)	.10 (0.46)	.08 (0.54)	.07 (0.55)
	FED	47 (3)	46 (5)	45 (9)	46 (7)			
Fat, %kcal	FAST	35 (8)	36 (5)	36 (4)	35 (7)	.09 (0.45)	.08 (0.38)	.08 (0.58)
	FED	35 (9)	36 (3)	37 (2)	36 (4)			
Energy, kcal/d	FAST	3472 (245)	3436 (249)	3425 (283)	3482 (275)	.20 (0.37)	.08 (0.45)	.09 (0.51)
	FED	3496 (368)	3432 (237)	3465 (232)	3472 (254)			
Total water intake, L/d	FAST	4.3 (0.5)	4.1 (0.4)	4.2 (0.3)	4.1 (0.7)	.06 (0.57)	.08 (0.35)	.20 (0.41)
	FED	4.2 (0.4)	4.3 (0.6)	4.2 (0.3)	4.1 (0.4)			

Abbreviations: ES, effect size; FAST, fasted state; FED, fed state; T0, 1 week before the start of Ramadan; T1, on the 15th day of Ramadan; T2, the day after the end of Ramadan; T3, 21 days after the end of Ramadan.

sleeping), sleep duration (period sleeping reported by subject), habitual sleep efficiency (ratio between time sleeping and time on bed), sleep disturbance (problems to sleep elected by subjects), sleeping medication (use of medication to sleep), and daytime dysfunction (intensity of diurnal somnolence).⁴⁴ The 7 domains are each scored from 0 (no difficulty for the component of interest) to 3 (very bad), with a total or global score ranging from 0 to 21 that reflects sleep quality and disturbances over a 1-month period. A better sleeper would have a lower global PSQI score, while higher score indicates poor sleep quality.

Statistical Analyses

The statistical analyses were conducted using the SPSS program version 16.0. Descriptive statistics are presented as mean (SD). Normal distribution of data was confirmed using the Shapiro–Wilk test. The effects of training times during RIF were evaluated using a mixed model of analysis of variance with 2 (groups: FAST and FED) $\times$ 4 (times: T0, T1, T2, and T3). If a statistically significant interaction or main effect was found, Bonferroni corrected post hoc tests were calculated to identify specific differences.

Effect sizes (ES) were determined from the output of analysis of variance by converting partial eta squared to Cohen *d*. In addition, within-group ES was computed using the following equation: ES = (mean post – mean pre)/SD. ES was considered trivial (<0.2), small (0.2–0.6), moderate (0.6–1.2), large (1.2–2.0), and very large (2.0–4.0).⁴⁵ All statistical analyses were computed at $P < .05$.

Results

All participants received treatments as allocated. Adherence to exercise was 100% for all participants. No training- or test-related injuries were recorded. No significant between-group differences were observed for any anthropometric measures, muscle CSA, and muscle strength measures at baseline (T0). Dietary intake, RPE values, and sleep quality in the FED and FAST groups were recorded at T0, T1, T2, and T3, and are summarized in Table 2. The changes in ES ranged from small to moderate for all measurements.

Anthropometric Tests

Anthropometric and body composition characteristics of participants are presented in Table 2. There were no significant group $\times$ time

interactions for anthropometric measures. A significant main effect of time was observed for body mass and body fat percentage. Both FAST and FED groups experienced significant decreases in body mass ($P = .001$; ES = 0.44), body mass index ($P = .001$; ES = 0.67), and body fat percentage ($P = .001$; ES = 0.91) during Ramadan, with no changes in lean body mass in either group ($P > .05$).

Hypertrophy and Maximal Strength

Group $\times$ time interactions are reported in Table 3 for measures of CSA and maximal strength. There were significant group $\times$ time interactions for 1RM test for the squats (1RM_{SQ}; $P = .001$; ES = 0.21) and deadlift (1RM_{DL}; $P = .04$; ES = 0.10) in favor of the FED group. Post hoc tests revealed significant increases in 1RM_{SQ} and 1RM_{DL} at T2 compared to T0 ($P < .05$) and at T3 compared to T0 and T1 ($P < .05$) for FED. No significant group $\times$ time effects were recorded for 1RM test for the bench press, cross-sectional area for quadriceps, and cross-sectional area for biceps brachii ($P > .05$).

Rating of Perceived Exertion

The RPE among FAST and FED groups at T0, T1, T2, and T3 is summarized in Figure 4, and significant group $\times$ time differences between groups for RPE were reported at T1 ($P = .02$; ES = 0.48) and T2 ($P = .03$; ES = 0.37). A significant main effect of time was observed among FAST group with increasing at T1 and T2, while RPE remained unchanged at all time points in the FED group.

The Pittsburgh Sleep Quality Index

Data of the PSQI (mean [SD]) among FAST and FED groups at T0, T1, T2, and T3 are reported in (Table 4).

There was significant group $\times$ time interaction for total score of PSQI, and a significant main effect of time was observed among both groups with decreasing of global PSQI at T1 ($P < .05$, $d = 6.54$) and at T2 ($P < .05$, ES = 6.54) compared to T0, and increasing of the total score for the PSQI during the 2 weeks after RIF. No significant differences between FAST and FED groups were found on any component.

Discussion

This study evaluated the effects of performing RT during Ramadan month on strength and hypertrophy in male athletes with previous

Table 2 Anthropometric and Body-Composition Characteristics of the FAST and FED Groups Measured During T0, T1, T2, and T3 (n = 20 per Group)

Variable	Group	Phase, mean (SD)				P (ES)		
		T0	T1	T2	T3	Time	Group	Group $\times$ time
Body mass, kg	FAST	79.12 (3.71)	76.06 (3.25)*	74.19 (3.20)*#	77.64 (4.52)\$	.001 (0.44)	.54 (0.05)	.09 (0.07)
	FED	80.05 (3.34)	79.08 (2.86)*	76.23 (3.03)*	78.04 (3.52)\$			
BMI, kg/m ²	FAST	24.26 (1.72)	23.6 (1.45)*	22.35 (1.34)*#	23.91 (1.19)\$	.001 (0.67)	.27 (0.03)	.13 (0.14)
	FED	24.72 (1.03)	24.24 (1.26)	23.31 (1.50)*	24.01 (1.61)\$			
Body fat, %	FAST	16.66 (1.25)	15.31 (1.60)*	13.53 (1.55)*#	14.84 (1.41)*\$	.001 (0.91)	.89 (0.00)	.58 (0.05)
	FED	16.20 (0.57)	15.31 (1.59)*	13.82 (1.65)*#	14.93 (1.47)*\$			
LBM, kg	FAST	66.78 (2.94)	65.84 (2.92)	64.78 (2.86)	65.70 (3.39)	.31 (0.31)	.10 (0.06)	.81 (0.01)
	FED	67.24 (1.88)	66.72 (2.45)	66.57 (2.59)	66.98 (2.91)			

Abbreviations: BMI, body mass index; ES, effect size; FAST, fasted state; FED, fed state; LBM, lean body mass; T0, 1 week before the start of Ramadan; T1, on the 15th day of Ramadan; T2, the day after the end of Ramadan; T3, 21 days after the end of Ramadan.

* $P < .05$ compared to T0. # $P < .05$ compared to T1. \$ $P < .05$ compared to T2.

Table 3 Maximal Strength (1RM) and Hypertrophy Changes of the FAST and FED Groups Measured During T0, T1, T2, and T3 (n = 20 per Group)

Variable	Group	Phase, mean (SD)				P (ES)		
		T0	T1	T2	T3	Time	Group	Group × time
1RM _{SQ} , kg	FAST	168.75 (13.13)	169.50 (11.85)	168.45 (11.54)	172.25 (11.55)*#	.001 (0.60)	.04 (0.12)	.001 (0.21)
	FED	169.65 (12.09)	170.78 (11.15)	174.23 (12.08)*	177.10 (12.69)*#			
1RM _{BP} , kg	FAST	106.01 (9.50)	110.40 (10.54)	109.25 (10.33)	120.10 (10.09)*	.001 (0.68)	.63 (0.006)	.38 (0.08)
	FED	105.50 (10.60)	107.35 (10.15)	109.50 (10.55)	116.75 (10.58)*			
1RM _{DL} , %	FAST	186.25 (8.13)	186.80 (8.88)	187.81 (9.38)	190.25 (9.88)*#	.001 (0.29)	.003 (0.12)	.04 (0.10)
	FED	187.50 (8.50)	190.85 (8.44)	195.40 (9.10)*	199.65 (8.39)*#			
CSA _{QC} , cm ²	FAST	69.95 (7.14)	68.90 (6.98)	68.15 (6.58)	71.15 (6.27)\$	.94 (0.01)	.18 (0.12)	.27 (0.03)
	FED	70.60 (6.20)	71.90 (6.80)	72.80 (6.70)	76.45 (6.64)*			
CSA _{BB} , cm ²	FAST	20.50 (2.10)	21.20 (2.42)	20.65 (2.45)	22.65 (2.69)*#\$	.001 (0.13)	.88 (0.42)	.29 (0.32)
	FED	20.90 (2.02)	20.45 (2.41)	20.91 (2.46)	22.15 (2.35)#			

Abbreviations: 1RM, 1-repetition maximum; BB, biceps brachii; BP, bench press; CSA, cross-sectional area; DL, deadlift; ES, effect size; FAST, fasted state; FED, fed state; QC, quadriceps; SQ, squat; T0, 1 week before the start of Ramadan; T1, on the 15th day of Ramadan; T2, the day after the end of Ramadan; T3, 21 days after the end of Ramadan.

* $P < .05$ compared to T0. # $P < .05$ compared to T1. \$ $P < .05$ compared to T2.

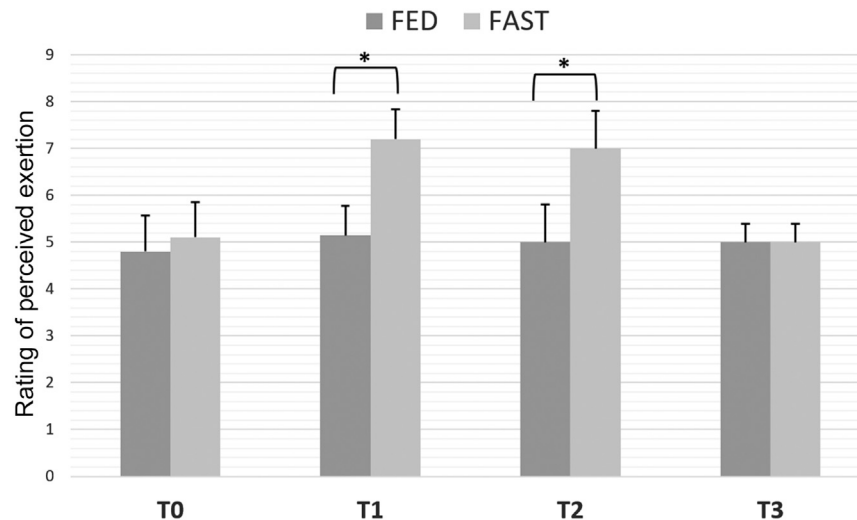


Figure 4 — Rating of perceived exertion among FAST and FED groups. FAST indicates fasted state; FED, fed state; T0, 1 week before the start of Ramadan; T1, the 15th day of Ramadan; T2, the day after the end of Ramadan; T3, 21 days after the end of Ramadan. * $P < .05$ significant differences between groups.

RT experience, and also compared the effects of practicing RT during RIF while still fasting (training before breaking fast) with RT in a nonfasting state (training after breaking fast). Our findings indicate that our first hypothesis was confirmed, and practicing RT during RIF has no negative effects on muscular performance in either group. However, our results did not support the second hypothesis. Thus, training in a FED positively affects only maximal strength with no significant development of muscle circumferences compared to the same total training volume while still fasting. Our findings regarding the lack of muscle hypertrophy in FAST have to be confirmed in future studies using more sophisticated measuring techniques (eg, magnetic resonance imaging). Our findings regarding the lack of muscle hypertrophy in FAST have to be confirmed in future studies using more sophisticated measuring techniques (eg, magnetic resonance imaging).

We studied the effects of RT in 2 well-controlled groups (FAST and FED) that had: (1) the same training conditions and

total volume training before and during Ramadan, (2) consumed no medications anabolic steroids, or ergogenic aids either before or during Ramadan, (3) maintained the same nutritional regimen with no significant effects on total energy and nutritional intake, and (4) reduced sleep schedule that was delayed only slightly (average ~1 h) but with no significant differences between groups.

Muscle Hypertrophy

There were neither adverse nor beneficial effects of the combination of RT and RIF on biceps and quadriceps muscle hypertrophy in the 2 study groups (as also reported in previous studies on the effects of RT during intermittent-fasting protocols). In fact, Moro et al⁴⁶ studied 34 males who performed RT for 8 weeks (3 resistance sessions/wk, 6–8 repetitions at 85–90% of 1RM) with restrictive feeding (16 h fasting/8 h fed, 3 days/wk) and reported no significant changes in CSA for muscles of the arms and thighs. However, other

Table 4 Sleep Quality Recorded of the FAST and FED Groups Measured During T0, T1, T2, and T3 (n = 20 per Group)

	Group	Phase, mean (SD)				P values (ES)		
		T0	T1	T2	T3	Time	Group	Group × Time
Sleep quality	FAST	1.2 (0.5)	2.4 (0.8)*	2.2 (0.7)*	1.3 (0.7)#,\$	.007 (0.48)	.06 (0.65)	.1 (0.56)
	FED	1.3 (0.7)	2.4 (0.7)*	2.3 (0.5)*	1.4 (0.6)#,\$			
Sleep latency, min	FAST	17.2 (5.4)	17.3 (4.7)	16.9 (8.4)	16.8 (6.4)	.06 (0.58)	.07 (0.47)	.2 (0.56)
	FED	16.9 (6.1)	17.5 (6.3)	17.1 (7.2)	17.1 (5.8)			
Sleep duration, h	FAST	7.9 (1.1)	6.3 (1.1)*	6.7 (1.2)*#	7.8 (1.2)#,\$	.002 (0.69)	.2 (0.65)	.06 (0.58)
	FED	8.1 (0.5)	6.1 (1.3)*	6.6 (1.5)*#	7.8 (1.5)#,\$			
Sleep efficiency, %	FAST	98.4 (3.5)	95.3 (8.2)*	95.2 (7.2)*	96.8 (6.5)#,\$	.005 (0.65)	.07 (0.37)	.1 (0.47)
	FED	97.7 (4.1)	94.8 (6.7)*	95.1 (6.8)*	97.1 (7.1)#,\$			
Sleep disturbance	FAST	0.7 (0.5)	0.7 (0.2)	0.6 (0.5)	0.6 (0.5)	.07 (0.45)	.08 (0.45)	.07 (0.55)
	FED	0.6 (0.3)	0.7 (0.5)	0.6 (0.4)	0.6 (0.4)			
Daytime dysfunction	FAST	0.4 (0.3)	0.4 (0.6)	0.4 (0.8)	0.5 (0.5)	.07 (0.38)	.2 (0.38)	.1 (0.56)
	FED	0.5 (0.6)	0.4 (0.7)	0.5 (0.6)	0.4 (0.3)			
Total score of PSQI	FAST	3.4 (2.7)	7.2 (2.3)*	7.5 (2.1)*	3.9 (2.7)#,\$	.003 (0.47)	.06 (0.43)	.08 (0.39)
	FED	3.8 (2.3)	7.7 (2.1)*	7.3 (2.6)*	4.1 (2.1)#,\$			

Abbreviations: ES, effect size; FAST, fasted state; FED, fed state; PSQI, Pittsburgh Sleep Quality Index; T0, 1 week before the start of Ramadan; T1, the 15th day of Ramadan; T2, the day after the end of Ramadan; T3, 21 days after the end of Ramadan.

* $P < .05$ compared to T0. # $P < .05$ compared to T1. \$ $P < .05$ compared to T2.

studies reported increases in muscle hypertrophy in males following RT in a time-restrictive feeding program.^{47,48} The discrepancy between our findings and the latter reports is likely related to conditions of fasting: We studied the effects of food and water deprivation for the entire month of Ramadan IF while the latter studies involved only 3 days per week of time-restrictive feeding.^{47,48} The continuous and prolonged fasting period during Ramadan decreases daily energy intake due to the reduction of meal frequency³⁰ which could hinder gains in muscle mass.

However, in this study, we found no significant differences in total energy and protein intake in the 2 study groups during the experiment; thus, the lack of muscle mass gain here may possibly be explained due to the changing of protein and glucose oxidation during RIF glycogen depletion in the liver. Muscle proteins (and not glucose) were used for the metabolism to sustain energy used by the other organs (heart and brain.) during starvation.^{49,50} Additionally, cumulative dehydration during fasting can also negatively impact muscle growth, as decreases in fluid consumption can alter serum concentrations of proteins and hormones associated with muscle protein synthesis.⁵¹

Muscle Strength

An important finding of our study is that there was no significant effect of RIF on 1RM test for the bench press in either the FAST or the FED study groups. However, there were slight improvements for 1RM_{DL} and 1RM_{SQ} in the FED group, with no adverse changes in the FAST group. These findings suggest that practicing RT during Ramadan can sustain or improve strength capacity whether it be practiced before or after breaking the fast, which is consistent with other reports indicating no adverse impacts of RIF on 1RM test for the bench press.⁵² Furthermore,²⁶ observed slight improvements in muscle strength capacity during RIF. The authors attributed this unchanged performance despite fasting to a constant total training volume and nutrient intake during RIF, as was the case for our study.

However, the improvement was observed just for the FED group with increases in 1RM_{DL} and 1RM_{SQ}. Choosing the optimal

training conditions, including timing of training, during Ramadan is of great interest to Muslim athletes. Several changes in physiological and psychological parameters during the daytime can impact exercise, including maximal strength performance. For example, the optimal time of day to conduct an acute high-intensity exercise session during Ramadan is after breaking the fast, as performance is negatively impacted in a fasting state (morning and evening sessions).⁵³

These findings may be due to exercise sessions at 9:00 PM which were performed under favorable circumstances that were similar to those before Ramadan where athletes were able to consume food and liquids before exercise sessions, allowing pre-exercise blood glucose concentration levels and fluid to be within the normal range, whereas training at 6:00 PM is not advisable as athletes have not consumed food and fluid for >10 hours and experience higher levels of fatigue.¹ Another explanation for the improvement only in the FED is that they reached their fatigue sensation later in comparison with the FAST group. To this point, in our study, RPE levels, recorded postexercise, were greater in fasting participants compared to the subjects in the fed state. This suggests that exercising while fasting may amplify the sensation of physical effort and fatigue independently of the physiological capacity of the muscles for the same training load.⁵⁴

Study Limitations

Our measurements were conducted on 2 different daytime points (in the late afternoon, just before breaking fast and in the evening, and after breaking the fast). Future studies should include additionally time points, such as a third time point in the morning to investigate the effects of training shortly after consuming the predawn meal of *suhur*.

Additionally, this study was conducted only with young males. Therefore, the observed findings cannot be translated to females. Future studies should examine this research question in female athletes as more women are now engaging in sports and exercise training.⁵⁵

Practical Applications

Muslim recreational weightlifters try to maintain their physical appearance during Ramadan despite the physiological and psychological challenges that occur during this month of daily fasting. Fitness instructors and personal trainers may need to implement strategies to prevent the negative effects of fasting on muscle hypertrophy and strength. Our findings suggest that it is feasible to maintain and increase maximal strength but not muscle circumference when exercise is performed in the evenings during the FED.

It is possible that the effects we observed may be even more pronounced in geographical areas where the fasting times are even longer (and conversely, less important in areas where duration of fasting is shorter). In addition, some non-Muslim weightlifters followed an intermittent fasting protocol as a nutritional method for muscle toning; however, trainers practicing RT should stress the importance of adopting an appropriate nutritional protocol to reach training goals.

Conclusions

Ramadan fasting and maintaining RT induce favorable changes in body composition (loss of fat and maintenance of lean tissue) without disturbing total energy and fluid intake. Furthermore, maintaining RT volume during RIF has no negative effect on strength and hypertrophy. Additionally, practicing after breaking fast leads to significant improvement on maximal strength. Thus, choosing the adequate time of training, practicing nutritional strategies, and maintaining the same training volume during Ramadan allow fasting recreational athletes to sustain and improve muscle performance.

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